OXFORD
INTERNATIONAL AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ____________________

Forename(s) ____________________

Candidate signature ____________________

INTERNATIONAL GCSE
BIOLOGY
Paper 1

Wednesday 8 May 2019   07:00 GMT   Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice
In all calculations, show clearly how you work out your answer.

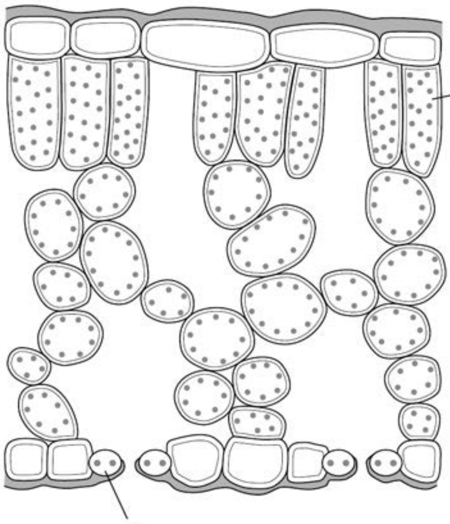
For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

Do not write outside the box

Answer **all** questions in the spaces provided.

0 1 . 1 **Figure 1** shows a section through a leaf.

Figure 1



Label **A** and **B** on **Figure 1**.

[2 marks]

Choose words from the box.

Guard cell	**Palisade mesophyll**	**Phloem**
Spongy mesophyll	**Xylem**	

0 1 . 2 Photosynthesis occurs in the leaf.

Complete the word equation for photosynthesis.

[2 marks]

$$\text{carbon dioxide + water} \xrightarrow{\text{light}} \underline{\qquad\qquad} + \underline{\qquad\qquad}$$

Choose words from the box.

energy	**glucose**	**hydrogen**	**oxygen**	**water vapour**

Do not write outside the box

0 1 . 3 Phloem and xylem transport substances to the leaf and from the leaf.

Draw **one** line from each tissue to the main substance it transports.

[2 marks]

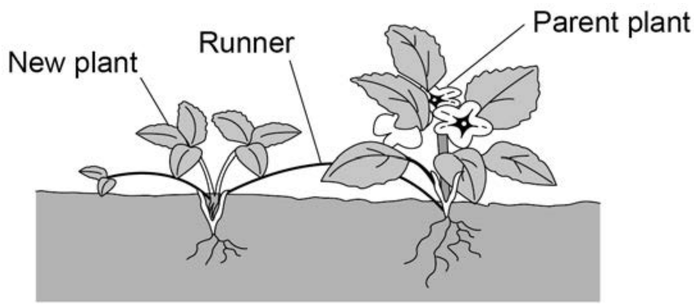
Tissue	**Substance**
	Carbon dioxide
	Oxygen
Phloem	Sugar
Xylem	Urea
	Water

Question 1 continues on the next page

0 1 . 4 **Figure 2** shows how new strawberry plants can grow.

The parent plant sends out runners from the stem. New plants can grow at points along the runner.

Figure 2



Which type of reproduction is shown by the runners of the strawberry plant?

[1 mark]

Tick (✓) **one** box.

Asexual ☐

Evolution ☐

Inheritance ☐

Sexual ☐

Do not write outside the box

0 1 . 5 A scientist can also produce new plants using small groups of cells from the original strawberry plant.

Name the process.

[1 mark]

____

____

0 1 . 6 Suggest **one** advantage and **one** disadvantage of using the process you named in question **01.5** compared with growing plants from seeds.

[2 marks]

Advantage ______________________________________

____

Disadvantage ____________________________________

____

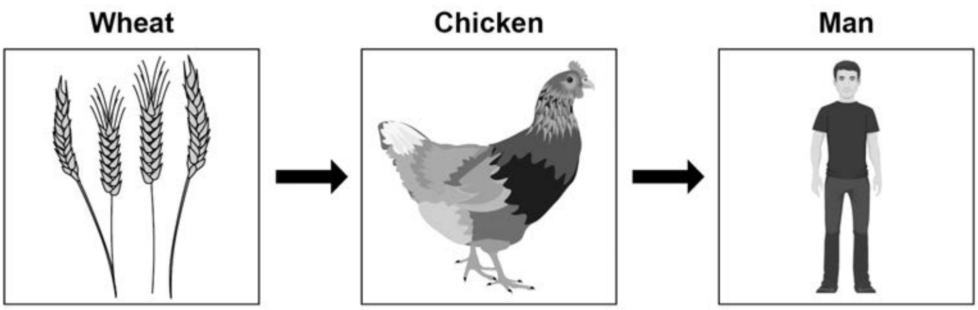
10

Turn over for the next question

Do not write outside the box

0 2 **Figure 3** shows a food chain.

Figure 3



Biomass	14 000 kg	700 kg	70 kg

0 2 . 1 What is the source of energy for the food chain in **Figure 3**?

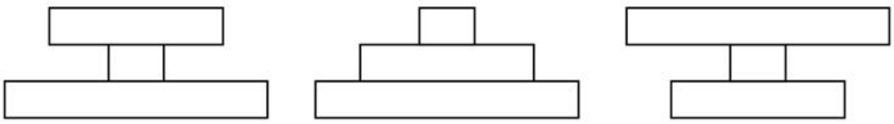
[1 mark]

____

0 2 . 2 Which diagram shows the pyramid of biomass for the food chain in **Figure 3**?

[1 mark]

Tick (✓) **one** box.



☐ ☐ ☐

Do not write outside the box

0 2 . 3 Some biomass is lost at each stage in the food chain.

In which **two** ways is biomass lost?

[2 marks]

Tick (✓) **two** boxes.

As food for the chicken ☐

As food for the man ☐

As waste from the chicken ☐

As wheat photosynthesises ☐

As wheat which is not eaten ☐

0 2 . 4 Calculate the total biomass lost in the food chain in **Figure 3**.

[1 mark]

____

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____

____

Total biomass lost = ______________________ kg

Question 2 continues on the next page

Do not write outside the box

Figure 4 shows two different ways of growing chickens.

Figure 4

Keeping chickens outside

Keeping chickens inside

0 2 . 5 Suggest **two** reasons why a farmer would keep chickens inside rather than outside.

[2 marks]

1 ______________________________

____

2 ______________________________

____

0 2 . 6 Suggest **one** reason why some people disagree with keeping chickens inside.

[1 mark]

____

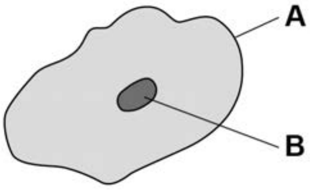
____

8

Do not write outside the box

0 3 **Figure 5** shows a human skin cell.

Figure 5



0 3 . 1 Label **A** and **B** on **Figure 5**.

[2 marks]

0 3 . 2 Cells are organised into tissues.

Each tissue has a particular function.

Draw **one** line from the type of tissue to the function of the tissue.

[2 marks]

Type of tissue	**Function**
	Covers some parts of the body
Glandular	Insulation
Muscular	Production of enzymes and hormones
	Movement

Question 3 continues on the next page

Do not write outside the box

If human heart tissue is damaged, it may **not** contract properly.

0 3 . 3 Why is it important for heart tissue to be able to contract?

[1 mark]

0 3 . 4 A group of cells known as the pacemaker controls the rate of heart contraction.

Where is the pacemaker located in the human heart?

[1 mark]

Tick (✓) **one** box.

Left atrium ☐

Left ventricle ☐

Right atrium ☐

Right ventricle ☐

Stem cells can be used to treat damaged tissues.

0 3 . 5 What is a stem cell?

[2 marks]

Do not write outside the box

Scientists have created stem cells from human skin cells.

0 3 . 6 Name **one** different source of human stem cells.

[1 mark]

0 3 . 7 Heart disease causes damage to heart tissue.

A damaged heart can be replaced using a donated heart.

It may be possible to replace heart tissue with tissue produced from the patient's own skin cells.

Suggest **two** reasons why heart tissue from stem cells might be better than a donated heart for treating someone with heart disease.

[2 marks]

1

2

0 3 . 8 Stem cells divide rapidly.

Suggest **one** possible risk of using stem cells in the human body.

[1 mark]

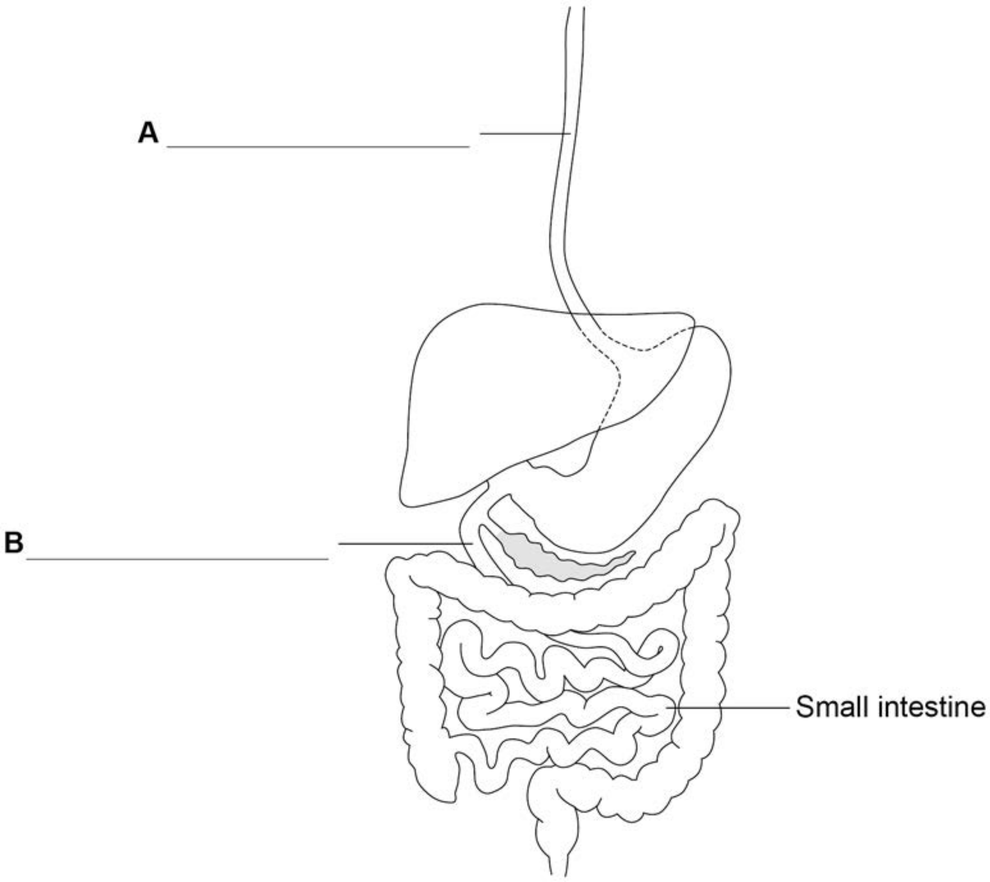
12

Turn over for the next question

Do not write outside the box

0 4 **Figure 6** shows part of the human digestive system.

Figure 6



0 4 . 1 Label **A** and **B** on **Figure 6** using the correct scientific terms.

[2 marks]

0 4 . 2 Meat is one source of protein in the diet.

Describe how:

- protein is broken down in the digestive system
- the products of protein digestion are absorbed into the body.

[4 marks]

Question 4 continues on the next page

Do not write outside the box

A diet high in meat has been linked to an increased risk of cancer of the colon (large intestine).

Doctors investigated the number of cases of colon cancer compared to daily meat intake.

Figure 7 shows the results of the study.

Figure 7

0 4 . 3 Draw a line of best fit on **Figure 7**.

[1 mark]

Do not write outside the box

0 4 . 4 The doctors concluded that there was a link between a diet high in meat and colon cancer.

Evaluate the doctors' conclusion.

Use information from **Figure 7**.

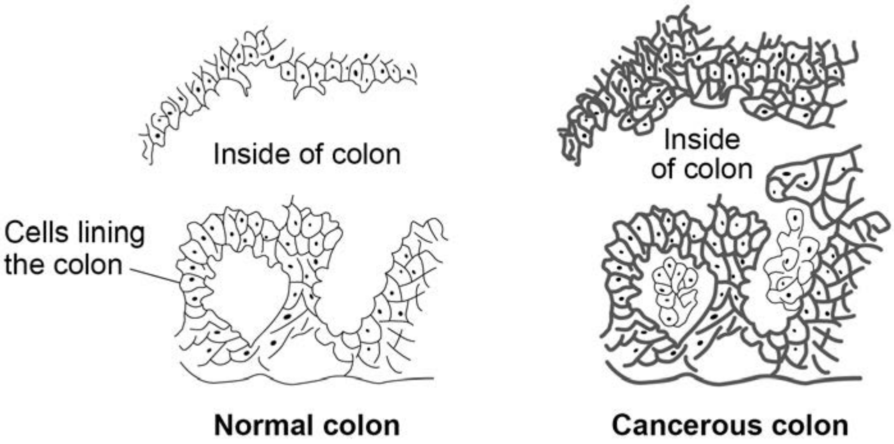
[3 marks]

Question 4 continues on the next page

0 4 . 5 Microscopes can be used to look at tissues to see if cancerous cells have developed.

Figure 8 shows sections through the lining of the colon from two people.

Figure 8



Describe **two** differences between the normal colon tissue and the cancerous colon tissue in **Figure 8**.

[2 marks]

1 ____

____

2 ____

____

Cancerous cells divide by mitosis.

0 4 . 6 Describe what happens when a cell divides by mitosis.

[2 marks]

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Do not write outside the box

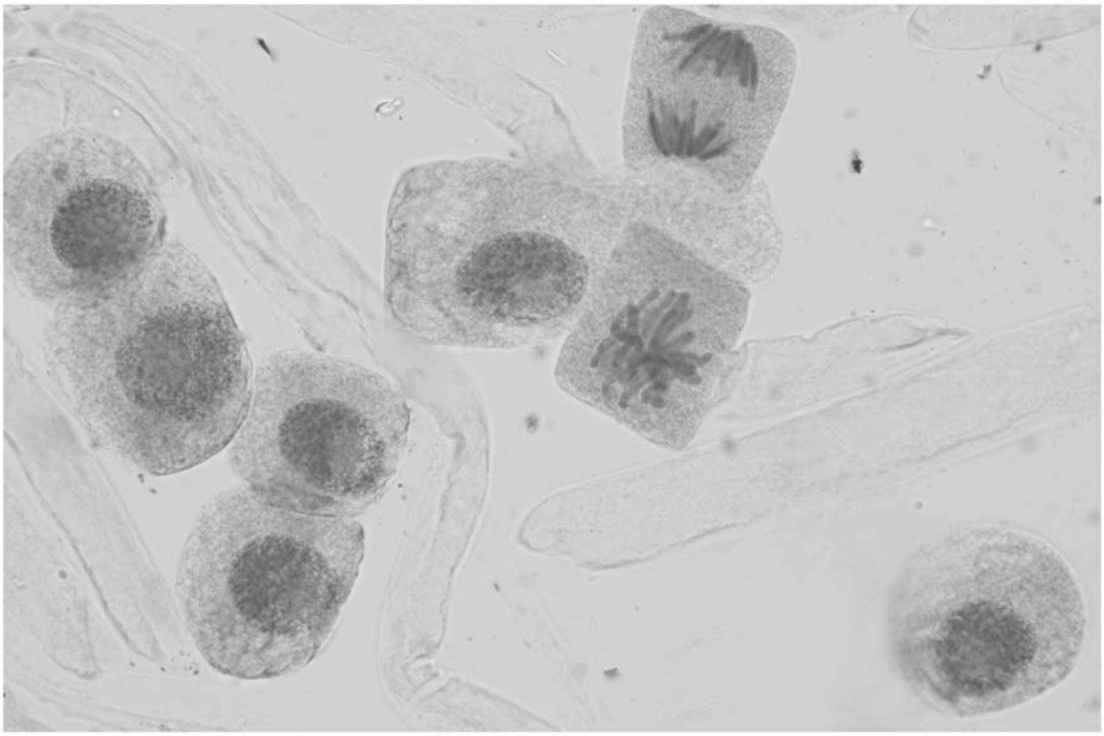
0 4 . 7 Mitotic index is a measure of the rate at which cells are dividing by mitosis.

Mitotic index is determined by examining cells under a microscope.

Chromosomes are visible in cells which are dividing.

Figure 9 shows cells dividing by mitosis.

Figure 9



Calculate the mitotic index of the cells in **Figure 9**.

Use the equation

$$\text{Mitotic index} = \frac{\text{number of dividing cells}}{\text{total number of cells}}$$

[2 marks]

Mitotic index = ____________________

Question 4 continues on the next page

Do not write outside the box

0 4 . 8 Scientists investigated the effectiveness of two drugs in the treatment of colon cancer.

The drugs stop mitosis and cause cells to die.

Each drug was tested on its own and combined together. The tests were carried out on four different types of colon tumour cells.

Table 1 shows the results.

Table 1

	Percentage (%) of cells that died		
Cell type	**Drug X**	**Drug Y**	**Drug X and Y**
1	52	64	78
2	59	55	66
3	62	52	72
4	42	41	44

Evaluate the use of drugs **X** and **Y** in the treatment of colon cancer.

Use calculations from the data in **Table 1** in your answer.

[4 marks]

20

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Turn over for the next question

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0 5 Sports scientists investigated the fitness levels of men.
The scientists tested 50 men at each of five different ages.

This is the method used.

1. Count the number of breaths taken in during one minute.
2. Repeat step 1 two more times and calculate a mean breathing rate per minute.
3. Swim 100 m as fast as possible.
4. Measure and calculate the mean breathing rate per minute after exercise.
5. Calculate the percentage increase in breathing rate.

Table 2 shows the results.

Table 2

Age in years	**Mean breathing rate per minute before exercise**	**Mean breathing rate per minute after exercise**	**Percentage increase in breathing rate (%)**
15	16	35	119
20	14	40	186
25	15	44	193
35	17	49	
45	20	50	150

0 5 . 1 Calculate the percentage increase in breathing rate for 35-year-olds.

[2 marks]

Percentage increase = ________________ %

0 5 . 2 Explain why the increase in breathing rate for each age group is given as a percentage.

[2 marks]

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0 5 . 3 Give **one** variable that was controlled during the investigation.

[1 mark]

0 5 . 4 The scientists used the mean percentage increase in breathing rate as a measure of fitness.

Suggest **one** other way the fitness of these men could have been determined.

[1 mark]

0 5 . 5 The body reacts to an increased demand for energy during exercise.

Explain how the body responds to meet the demand for energy during exercise.

Do **not** refer to changes in breathing rate.

[4 marks]

Question 5 continues on the next page

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0 5 . 6 The lower the percentage increase in breathing rate, the fitter the men are.

Men aged 25 years = 193% increase in breathing rate.
Men aged 20 years = 186% increase in breathing rate.

The scientists concluded that men aged 20 were significantly fitter than those aged 25.

The scientists used a statistical test (**S**) to see if their conclusion was justified.

This test shows if there is a **significant** difference between two sets of results.

The test is calculated using the equation

$$\mathbf{S} = \frac{[\text{percentage increase in breathing rate at age A} - \text{percentage increase in breathing rate at age B}]}{\sqrt{2}}$$

Calculate **S** for ages 25 and 20 where:

- Age A = 25
- Age B = 20

[2 marks]

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S = ______________________

Do not write outside the box

0 5 . 7 The value of **S** is compared to a critical value.

If **S** is **greater** than the critical value then the results are significantly different.

Table 3 shows the critical values.

Table 3

Number of age groups	Critical value
2	12.71
3	4.30
4	3.18

What can you conclude from the data on percentage increase in breathing rate for 20 and 25 year old men?

Give a reason for your answer.

[1 mark]

13

Turn over for the next question

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0 6 Genes control the characteristics of an organism.

0 6 . 1 What is a gene?

[1 mark]

Early onset Alzheimer's (EOA) is an inherited disorder which causes severe memory loss.

In one form of EOA, a mutation leads to the production of a changed amyloid beta (AB) protein.

The mutation leads to a changed DNA sequence in the AB gene.

0 6 . 2 Explain how the mutation could result in a change to the AB protein.

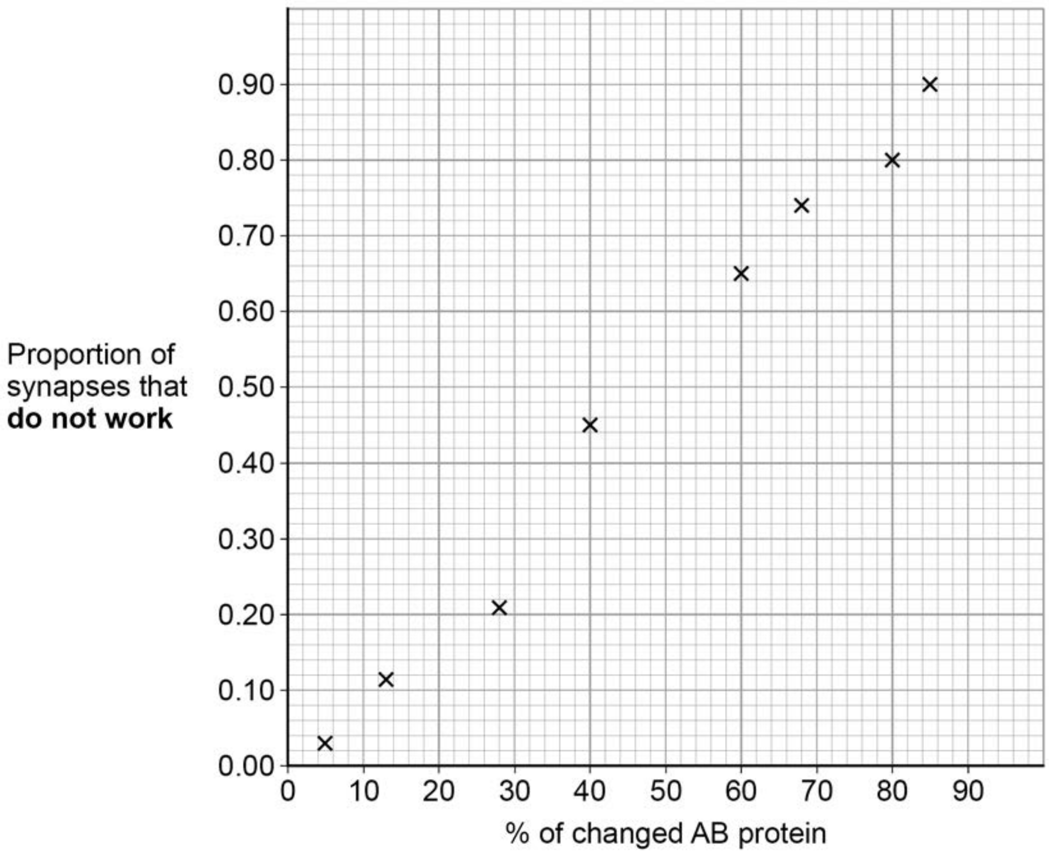
[3 marks]

Do not write outside the box

AB protein is important for the synapses in the nervous system to work correctly.

Figure 10 shows how the percentage of changed AB protein affects the proportion of synapses that do **not** work correctly.

Figure 10



0 6 . 3 The proportion of synapses that **do work** decreases between 40% and 80% changed AB protein.

Calculate the percentage decrease in the proportion of synapses that **do work** between 40% changed AB protein and 80% changed AB protein.

[3 marks]

Percentage decrease = ____________________ %

Question 6 continues on the next page

Turn over ►

When synapses do not respond to a stimulus for a long period of time, the nerve cells connected to the synapse die.

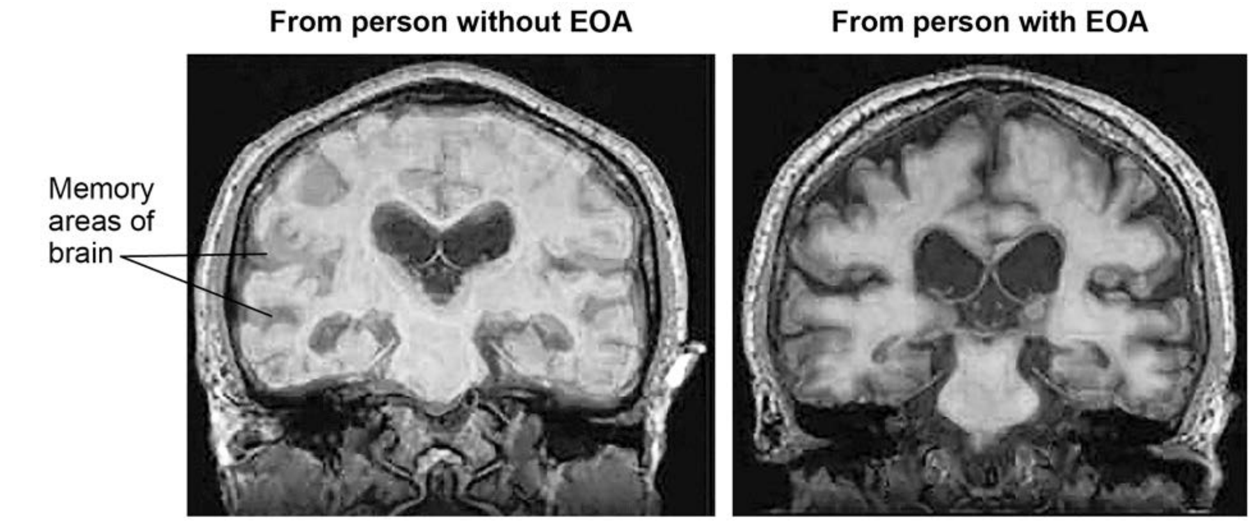
Figure 11 shows scans of two brains.

Figure 11

Key

Areas of brain with changed AB protein

Areas of brain with no changed AB protein



0 6 . 4 Suggest why loss of memory occurs in a person with EOA.

Use information from **Figure 10**, on page 25, and **Figure 11** above.

[4 marks]

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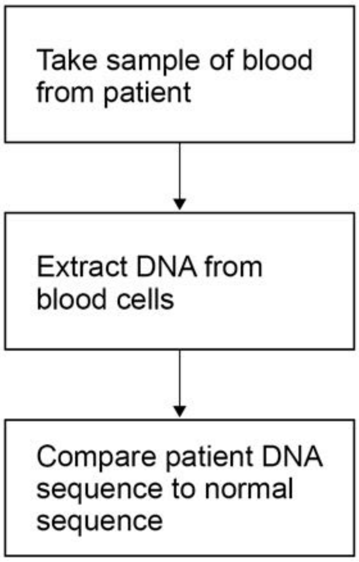
People at risk of developing EOA can be offered genetic screening.
Genetic screening shows if the mutation for AB is present or not.

Figure 12 outlines the process involved in genetic screening.

Figure 12



0 6 . 5 Why is the patient's DNA compared to normal DNA?

[1 mark]

____

____

0 6 . 6 People who have EOA in their family can choose whether or not to have genetic screening.

Suggest **one** advantage and **one** disadvantage of genetic screening for the family.

[2 marks]

Advantage ____

____

Disadvantage ____

____

14

Turn over for the next question

0 7 Development around a lake led to an increase of sewage.

The biodiversity in the lake rapidly declined.

0 7 . 1 Explain why an increase in sewage entering the lake reduces the biodiversity in the lake.

[5 marks]

A study was carried out to investigate the quality of the lake water at different times of the year.

This is the method used.

1. Choose six sites on the lake and collect water samples at a depth of 5 m.
2. Record dissolved oxygen concentration and water temperature.
3. Store water samples at -4 °C.
4. Measure nitrate ion concentrations.
5. Repeat steps 1 – 4 at various times in the year.

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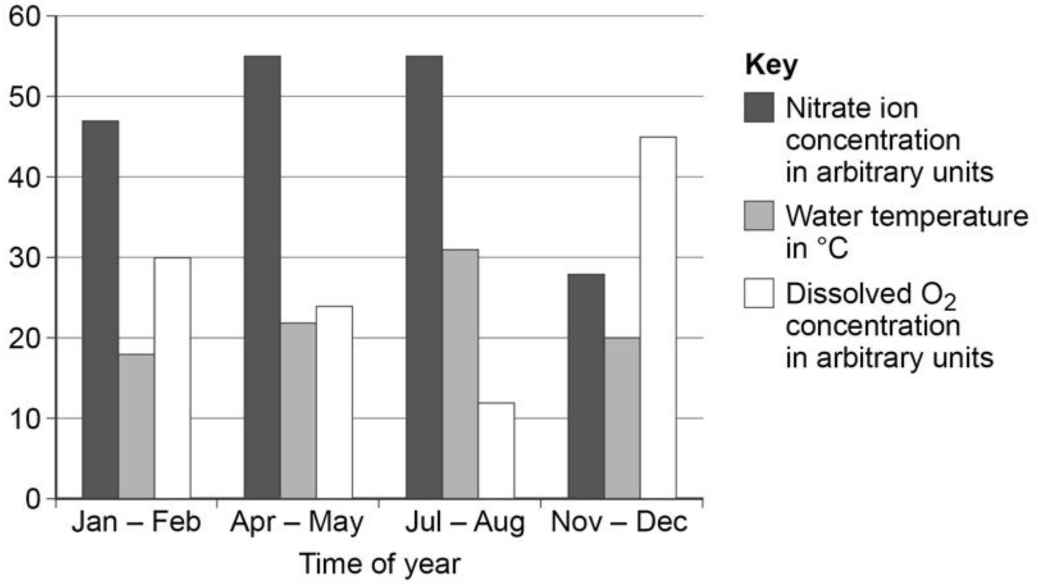
Figure 13 shows the results.

Figure 13



0 7 . 2 Suggest why the water samples were stored at -4 °C.

[1 mark]

____

____

0 7 . 3 Describe **two** trends in **Figure 13**.

Use data in your answer.

[2 marks]

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Turn over ►

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0 7 . 4 A sample of lake water was collected in October.

The sample was evaporated to dryness in a conical flask.

The mass of solids remaining can be used to calculate the concentration of dissolved solids in the lake water.

Table 4 shows the results.

Table 4

Volume of water in dm^3	0.5
Mass of empty flask in g	145.988
Mass of flask and water in g	651.341
Mass of dried flask in g	146.076

Calculate the concentration of dissolved solids in the water sample from the lake.

Give your answer in mg per dm^3

[3 marks]

Concentration of dissolved solids = ____________________ mg per dm^3

0 7 . 5 July is summer at the lake.

Suggest why the dissolved oxygen concentration is lowest at this time.

[2 marks]

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____

13

END OF QUESTIONS

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